Project Development Phase

Data Preprocessing Steps & System Dashboards

Metric	Details
Date	08 July 2026
Team ID	K. Bhargav
Project Name	Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease
Maximum Marks	4 Marks

Data Preprocessing Steps

Step 1: Import the Dataset

- Imported the clinical Heart Disease dataset (`Heart_new2.csv`) into Tableau Public and the local Python environment.
- Verified that all 18 records and fields (`HeartDisease`, `BMI`, `Smoking`, `AlcoholDrinking`, `Stroke`, `AgeCategory`, `Race`, `Diabetic`, etc.) were imported successfully.
- Confirmed that the ~4,494 patient observation records were fully loaded and ready for preprocessing.

Step 2: Check for Missing Values

- Examined the raw dataset for null, blank, or NaN (Not a Number) values.
- Verified that critical clinical fields such as `HeartDisease` (Target Variable), `BMI`, `AgeCategory`, and `Stroke` contained valid, non-empty information.
- Confirmed the dataset integrity, validating that zero null fragments required imputation, ensuring highly accurate epidemiological analysis.

Step 3: Remove Duplicate Records

- Checked the dataset for overlapping or identical patient observation rows that could skew clinical counts.
- Filtered and removed exact duplicate entry rows to prevent inaccurate patient demographics counts and incorrect medical insights.
- Ensured that every unique patient profile was represented only once in the final analytical matrix.

Step 4: Validate Data Types

Ensured that each attribute was mapped to the correct computational data type for Tableau processing:

- `HeartDisease` → Categorical (String: Yes/No)
- `BMI` → Number (Decimal / Float64)
- `SleepTime`, `PhysicalHealth`, `MentalHealth` → Number (Integer)
- `AgeCategory` → Categorical (String Bracket)
- `Sex`, `Race`, `GenHealth` → Categorical
- `Smoking`, `AlcoholDrinking`, `Diabetic`, `Stroke`, `Asthma`, `KidneyDisease`, `SkinCancer` → Categorical (Boolean String: Yes/No)

Step 5: Verify Data Consistency

- Checked for invalid or unrealistic values across continuous variables (e.g., verifying that `BMI` values fell within the realistic human scale of 16.6 to 54.75, and `SleepTime` contained no negative hours).
- Standardized categorical text formats (e.g., ensuring "Yes" and "No" labels were capitalized consistently) to prevent grouping errors inside visual charts.

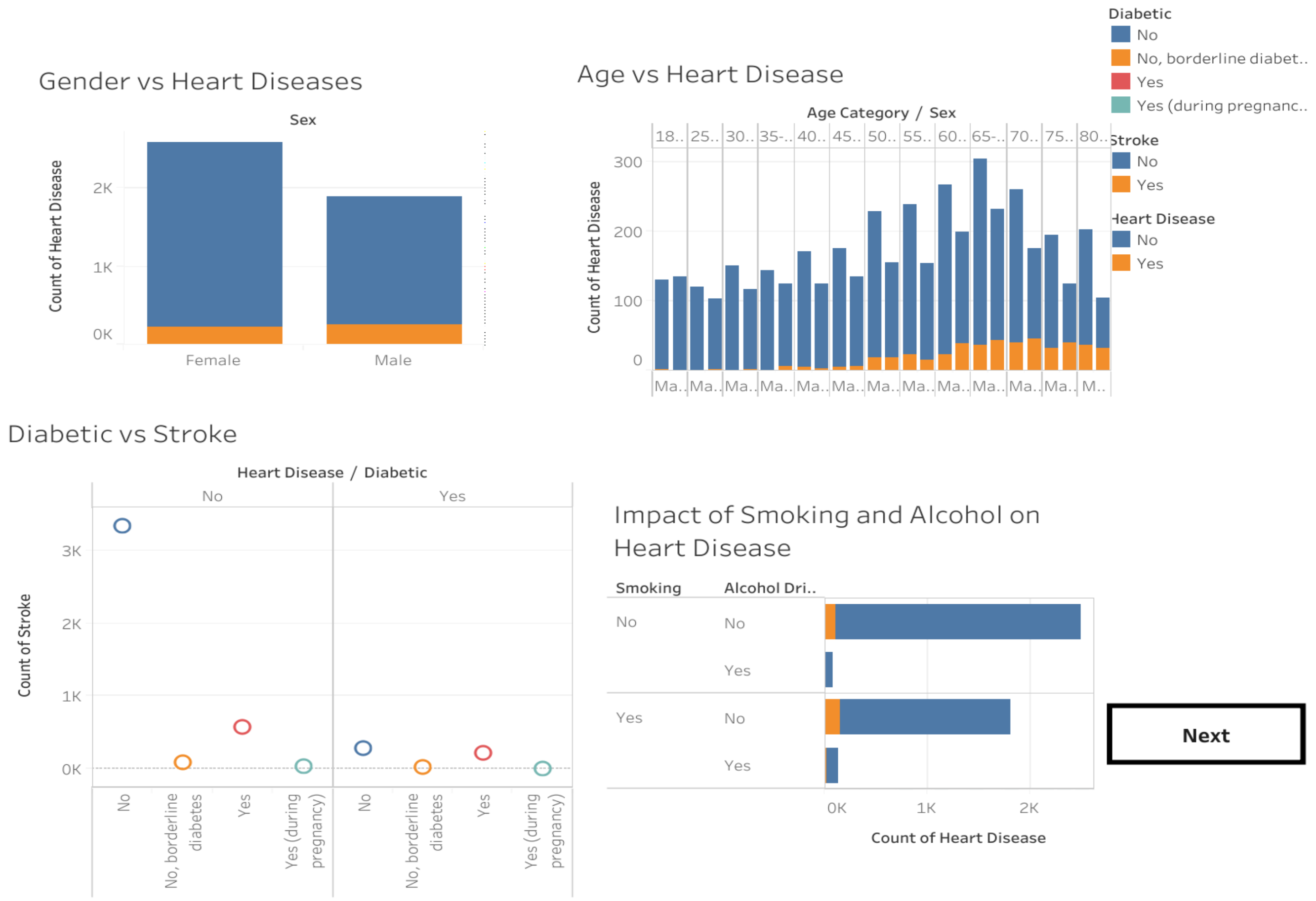
S.No.	Primary Business Question (Clinical Inquiry)	Target Dashboard / Component	Visual Mechanism & Chart Type
1	How do distinct age cohorts and biological gender profiles map to overall heart disease diagnostic tracking totals?	**Dashboard 1:** Baseline Demographics Matrix	**Stacked Bar Charts:** Tracks patient volumes across consecutive age brackets (18-24 up to 80 or older) partitioned cleanly by Male and Female categories.
2	In what ways do behavioral inputs like active smoking habits and heavy alcohol consumption compound to trigger critical cardiac incidents?	**Dashboard 2:** Lifestyle Indicators	**Cross-filtered Matrix Bars:** Dynamically isolates and compares joint occurrences of concurrent smokers and alcohol users against affirmative disease cases.
3	Can pre-existing conditions like Diabetes Mellitus and histories of prior Stroke serve as major indicators of cardiovascular complications?	**Dashboard 1 & 3:** Co-morbidity Mapping	**Scatter & Dot Plots:** Tracks varying diabetic phases (No, Borderline, Yes, Yes during pregnancy) mapped directly to absolute historical stroke event counters.
4	How do secondary chronic ailments (Asthma, Kidney Disease, Skin Cancer) overlap with cardiovascular and stroke records?	**Dashboard 2:** Secondary Ailment Profiling	**Segmented Column Charts:** Accurately reflects patient counts across multi-disease condition matrices to isolate severe secondary medical risks.
5	What is the epidemiological distribution and percentage breakdown of heart conditions across different ethnic demographics?	**Dashboard 2:** Epidemiological Distributions	**Pie Charts:** Renders percentage distribution slices (e.g., White: 75.57%, Black: 17.81%, Hispanic: 3.73%, Asian: 2.07%) to capture demographic variances.
6	How do qualitative, self-reported general health ratings correlate with active vs. sedentary physical activity metrics?	**Dashboard 2:** General Health Metrics	**Bubble Packing Charts:** Displays absolute volumes (Poor: 92, Fair: 140, Very Good: 67, Good: 172) alongside binary distribution charts for physical activity.
7	How do severe body mass vectors (BMI) distribute across age categories and diabetic status to form high-risk metabolic clusters?	**Dashboard 3:** Deep Metabolic Risk Grid	**Gradient Tree-Maps:** Visualizes varying intensities of Average BMI ranging from a safe 23.91 to a severe 42.78 using an intuitive heat-mapped color scale.

Final System Dashboards

Below are the verified interactive dashboard layouts successfully generated from the preprocessed data, ready for Flask web embedding:

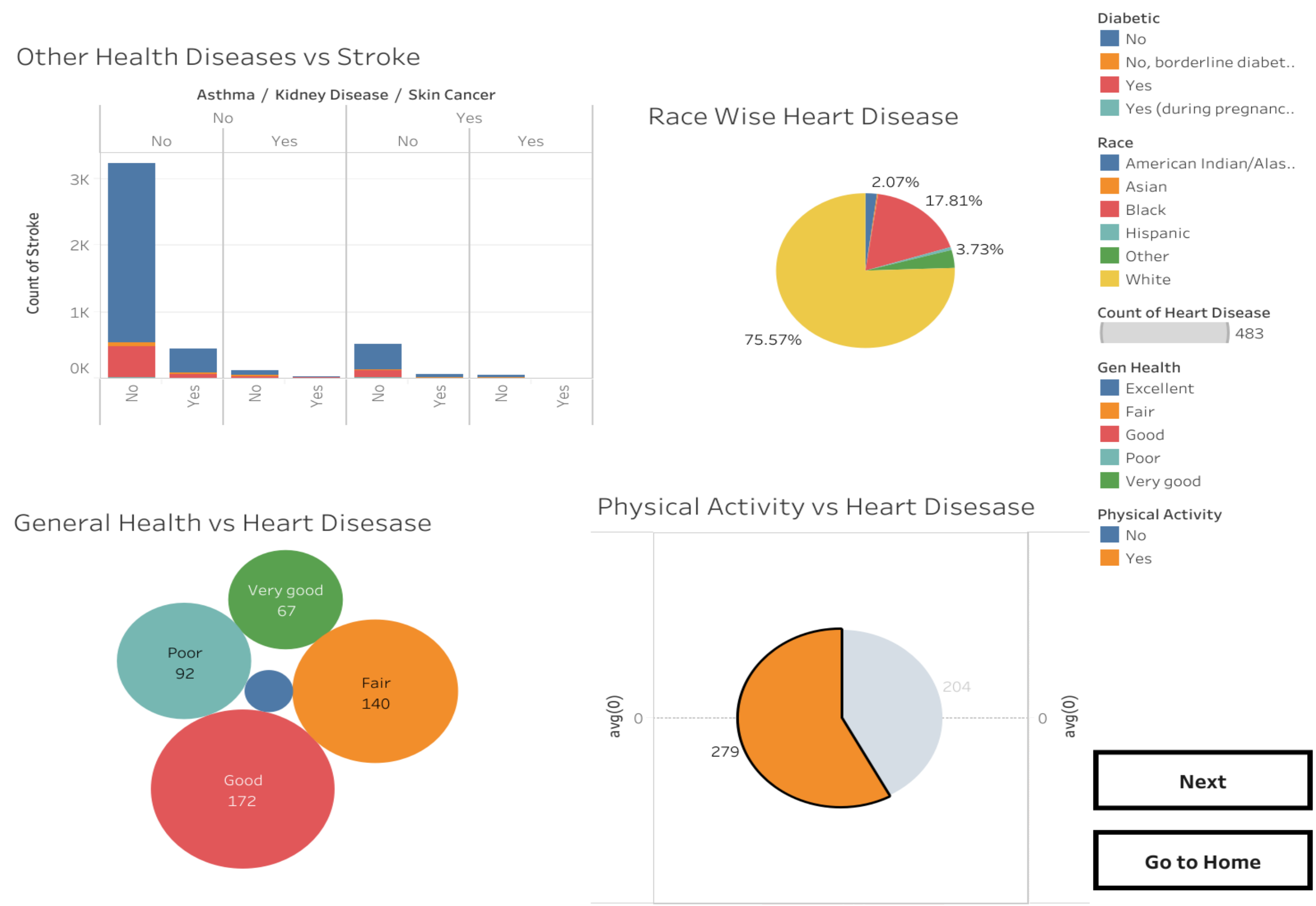
Dashboard 1: Baseline Demographics & Clinical Factors Matrix

Visualizes patient case volume counts segmented directly across age categories and biological sex profiles



Dashboard 2: Secondary Co-morbidities & Lifestyle Indicators

Maps qualitative user traits and epidemiological proportions, tracking joint occurrences of Asthma, Kidney Disease, Skin Cancer, and Race-wise dis



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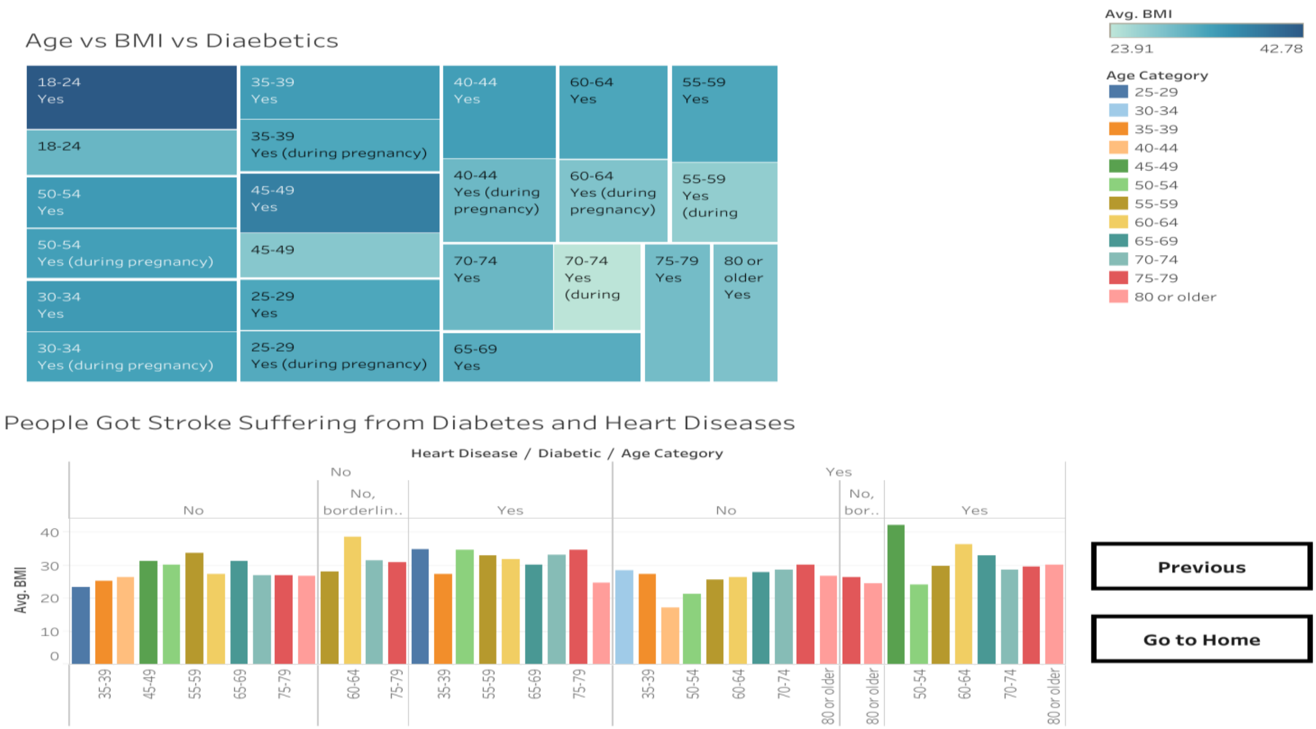
Dashboard 3: Deep Metabolic Risk Analysis Grid

A specialized tree-map layout showing Age vs. BMI variables to isolate severe body mass vectors ranging between baseline and advanced risk thresholds



Prepared as part of the Predictive Risk Clustering and Visual Informatics for Cardiovascular Disease capstone project.